CAA APPROVED FLIGHT MANUAL & PILOT'S OPERATING HANDBOOK

Airtourer T6

> **NOTE:** This document must be carried in the aircraft at all times.

FLIGHT MANUAL APPROVAL

This aircraft must be operated at all times in accordance with the approved data and limitations contained in this manual.

Field	Detail
Aircraft Manufacturer	Aero Engine Services Limited
Certification Categories	Semi-Acrobatic & Acrobatic
Certificate of Type Approval	New Zealand № A-8
Applicable Serial Numbers	A542, A546, A552, A553, B557–B567, B569–B571, B573–B575, B577–B580
Document Number	AIR 2792
Aircraft Serial Number	B571
Nationality & Registration	VH-BVV
Approved by	Civil Aviation Authority
Date	13 JAN 2003

This manual has been approved by the Civil Aviation Authority of New Zealand and is the Flight Manual referred to in the Type Certificate Data Sheet.

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GENERAL AMENDMENT RECORD SHEET

Amendment №	Description	Signature & Date of Incorporation
0	Complete re-issue of manual previously identified as AIR 72	

> **NOTE:** Amendment numbers may not always be consecutive.

INTRODUCTION

This Flight Manual/POH applies only to the particular aircraft identified by the serial number and registration marking on the Approval Page and contains the airworthiness limitations and essential operating data for this aircraft.

Flight Manual Supplements may be required when the aircraft is modified by the incorporation of any Supplemental Type Certificates or other modifications. These supplements will be provided by the STC holder or the design organisation involved. Approved supplements may be found in Section 9 of this manual.

This Flight Manual shall be carried in the aircraft on all flights. The Pilot-in-command shall comply with all requirements, procedures and limitations with respect to the operation of the aircraft set out in the Flight Manual.

Amendments will be approved by CAA as necessary and will take the form of replacement pages with changes to the text indicated by a vertical line in the margin together with the amendment number at the bottom of the page. It is the Certificate of Registration holder's responsibility to ensure that all required amendments are incorporated in this manual, and to ensure that details of the amendment are entered and certified for on the Amendment Record Sheet.

Other than for approved amendments, no entries or endorsements may be made to this Flight Manual.

LOG OF EFFECTIVE PAGES

Section	Pages	Amendment №	Date Issued
Intro	(i) through (v) inclusive	0	15 August, 2002
1	1-1 through 1-12 inclusive	0	15 August, 2002

Section	Pages	Amendment №	Date Issued
2	2-1 through 2-9 inclusive	0	15 August, 2002
3	3-1 through 3-6 inclusive	0	15 August, 2002
4	4-1 through 4-18 inclusive	0	15 August, 2002
5	5-1 through 5-5 inclusive	0	15 August, 2002
6	6-1 through 6-5 inclusive	0	15 August, 2002
7	7-1 through 7-8 inclusive	0	15 August, 2002
8	8-1 through 8-9 inclusive	0	15 August, 2002
9	9-1 through 9-2 inclusive	0	15 August, 2002

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SECTION 1 — GENERAL

1.1 Introduction

This CAA Approved Flight Manual/Pilots Operating Handbook consists of an introductory section and nine additional numbered sections and has been prepared to comply with the GAMA Handbook Specification № 1. Sections 1, 2, 3, 4, 5, 6 and 9 comprise the Approved Flight Manual. These sections, together with the remainder of the manual, comprise the Pilots Operating Handbook.

The handbook is not intended to be a guide for basic flight instruction or as a training manual. It may be used for operational purposes only if kept in a fully amended state. It contains all the information considered necessary to safely operate the aircraft.

1.1.1 Pilot's Operating Handbook (POH)

1. Approved Flight Manual — Sections 1, 2, 3, 4, 5, 6 and 9.

2. POH — The POH consists of the sections that comprise the Approved Flight Manual as well as the additional information presented in Sections 7 and 8.

3. Supplements — Self contained supplements are provided in Section 9 of the POH to provide details and procedures associated with the fitment of specified optional and special purpose equipment or to cover specific types of operations that the aircraft may be approved for.

4. Amendments — Amendments to any section of the POH is to have an amendment number allocated to it.

> **NOTE:** *All amendments are to be incorporated as soon as possible after their receipt and details entered into the appropriate amendment record sheet.*

1.1.2 Definitions

> **WARNING** — *Operating procedures, techniques, etc. which if not followed correctly, may result in personal injury or death.*

> **CAUTION** — *Operating procedures, techniques, etc. which if not strictly observed, may result in damage to the aircraft or to its installed equipment.*

> **NOTE** — *Operating procedures, techniques, etc. which it is considered essential to highlight.*

1.2 General Description

1.2.1 General Description of the Aircraft

The Airtourer T6 aircraft is a low wing, fixed tricycle undercarriage, single engine, two seat light aircraft. The fuselage consists basically of two main sections — the rear fuselage and the front fuselage — joined together to form a single unit. The front fuselage extends from the firewall to the rear of the cockpit and incorporates the cockpit, windshield and sliding canopy. The rear fuselage extends from the rear of the cockpit to the tail cone. The fuselage is of conventional rivetted aluminium alloy sheet metal construction.

1.2.2 Engine & Propeller

A four cylinder, horizontally opposed, air cooled, normally aspirated Textron Lycoming O-320 engine is installed, rated at 150 or 160 BHP. The engine drives a two blade Hartzell constant speed propeller.

1.2.3 Seating Capacity

Two seats are provided, side by side, and enclosed by a fixed windscreen and a sliding perspex canopy.

1.2.4 Wing

The wing is of single-spar construction, secured to the fuselage by means of fore and aft pins passing through the main spar from the spar attachment frame below the seats, and by spanwise bolts passing through the fuselage lower longerons and the angle brackets on the rear spar. Full span flapperon/ailerons are fitted and operated in conjunction with each other as ailerons with stick side movement, as flaps by means of the flap lever, with differential movement between flaps and ailerons.

1.2.5 Undercarriage

Tricycle type undercarriage is fitted. Nose-wheel steering is achieved by direct linkage with rudder pedals. The main undercarriage is attached to the wing ribs and consists of two leaf spring type shock absorbing legs of fail/safe design and low pressure wheels. The nose-wheel leg is a coil spring automotive hydraulic shock absorber assembly fitted with a low pressure wheel. Dunlop hydraulic disc brakes are fitted, and are operated by individual hand pull levers within easy reach of the pilot. A centrally mounted park brake selector can be used by either pilot to maintain brake pressure after individual levers have been released.

1.2.6 Empennage

The empennage consists of a single fin and aero-dynamically balanced rudder mounted wholly above the fuselage, and a single piece tailplane with full span elevator.

1.2.7 Fuel System

The fuel system consists of a rubber bag fuel tank located in the wing centre section beneath the pilots seats. A filler neck is provided on the starboard side of the fuselage above the wing. The tank is vented via a small air scoop located mid way down the starboard side of the rear fuselage. A sediment drain is located underneath the fuselage on the port side just near the inboard end of the port flapperon. The fuel tank has a total capacity of 131.8 litres.

1.3 General Specifications

Parameter	Value
Length	6705.6 mm [22.00']
Height	2133.6 mm [7.0']
Wing Span	7924.8 mm [26.0']
Wing Mean Aerodynamic Chord	1406.7 mm [4.615']
Wing Area	11.148 m^2 [120 ft^2]
Aspect Ratio	5.7
Dihedral	6° 4'
Undercarriage Track	2895.6 mm [9.5']
Total Fuel Capacity	131.8 lt
Useable Fuel Capacity	130.5 lt
Max. Take-Off Weight (Normal Category)	862 kg [1900 lb]
Max. Acrobatic Weight	816 kg [1800 lb]
Wing Loading (1900 lb)	77.3 kg/m^2 [15.83 lb/ft^2]
Power Loading (1900 lb/150 BHP)	12.7 lb/HP
Power Loading (1900 lb/160 BHP)	11.9 lb/HP
Maximum Baggage	45 kg [100 lb]
Take-Off Flap Setting	8°
Landing (Max) Flap	15°

1.4 Symbols, Abbreviations & Terminology

1.4.1 General Symbols and Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
A	Ampere	In	Inch, inches
AGL	Above Ground Level	In Hg	Inches of mercury
AMSL	Above Mean Sea Level	In lbs	Inch pounds
AVGAS	Aviation Gasoline	ISA	International Standard Atmosphere
BHP	Brake Horse Power	kg	Kilogram
CASA	Civil Aviation Authority	kHz	Kilohertz
CAO	Civil Aviation Order	kts, K	Knots
CAR	Civil Aviation Rules	kPa	Kilopascals
°C	degrees Celsius	kW	Kilowatts
CHT	Cylinder Head Temperature	lt	Litre, litres
cm	Centimetre	lb	Pound, pounds
DC	Direct Current	LH	Left hand
E	East	LHS	Left hand side
EMERG	Emergency	m, M	Metre
FAA	Federal Aviation Administration	m^2, M^2	Square metre
°F	degrees Fahrenheit	mA	Milli ampere
FAR	Federal Aviation Regulation (USA)	MAC	Mean Aerodynamic Chord
ft	Foot, feet	MAN	Manual
ft/min	Feet per minute	max	Maximum
g	Acceleration due to gravity	MCP	Maximum Continuous Power
GAL	Gallons	MHz	Megahertz
GAMA	General Aviation Manufacturers Assoc	mm	Millimetre

Abbrev.	Meaning	Abbrev.	Meaning
hPa	Hectopascal	min	Minimum or minute
ICAO	International Civil Aviation Org	N	North
ICO	Idle Cut Off	nm	Nautical mile
IFR	Instrument Flight Rules	PAX	Passenger
IMC	Instrument Meteorological Conditions	POH	Pilots Operating Handbook
PPH	Pounds per hour	STBY	Standby
PROP	Propeller	STC	Supplemental Type Certificate
psi, PSI	Pounds per square inch	TBO	Time between overhauls
PWR	Power	T/O	Take Off
QTY	Quantity	US	United States (of America)
qts	Quarts	U/S	Unserviceable
RH	Right Hand	USA	United States of America
RHS	Right Hand Side	V	Volts
RPM	Revolutions per minute	VFR	Visual Flight Rules
S	South	VHF	Very High Frequency
SAE	Society of Automotive Engineers	VMC	Visual Meteorological Conditions
sec	Seconds	W	West

1.4.2 General Airspeed Terminology and Symbols

Symbol	Definition
CAS	*Calibrated Airspeed:* the indicated speed of an aircraft corrected for position and instrument error. Equal to true airspeed in standard atmosphere at sea level.
KCAS	Calibrated Airspeed expressed in "Knots".
IAS	*Indicated Airspeed:* the speed of an aircraft as shown on the airspeed indicator. IAS values in this manual assume zero instrument error.
KIAS	Indicated Airspeed expressed in "Knots".

Symbol	Definition
TAS	*True Air Speed:* the airspeed of an aircraft relative to the air through which it passes.
TOSS	*Take-Off Safety Speed:* the minimum airspeed to which the aircraft must be accelerated during the take-off manoeuvre. The aircraft must have achieved this speed by 50 feet.
Va	*Manoeuvring Speed:* the maximum speed at which application of full available aerodynamic control will not damage or overstress the aircraft.
Vfe	*Maximum Flap Extended Speed:* the highest speed permissible with wing flaps in a prescribed extended position.
Vne	*Never Exceed Speed:* the airspeed that may not be exceeded at any time.
Vno	*Maximum Structural Cruising Speed:* the speed that should not be exceeded except in smooth air and then only with caution.
Vs	*Stalling Speed:* the speed at which the aircraft stalls in the nominated configuration.
Vso	*Stalling Speed:* the speed at which the aircraft stalls in the landing configuration.
Vx	*Best Angle-of-Climb Speed:* the airspeed which delivers the greatest gain of altitude in the shortest possible horizontal distance.
Vy	*Best Rate-of-Climb Speed:* the airspeed which delivers the greatest gain in altitude in the shortest possible time.
V_APP	*Approach Speed:* the airspeed chosen to ensure that adequate control will exist under all conditions, including turbulence, to carry out a normal flare and touchdown. It is the speed required at 50 feet during the approach.

1.4.3 Meteorological Terminology

ISA — *International Standard Atmosphere* in which the air is a dry perfect gas; temperature at sea level is 15°C; pressure at sea level is 1013.25 hPa; temperature gradient from sea level to 36,089 feet is -1.98°C per 1000 feet.

OAT — *Outside Air Temperature:* the outside free air static temperature.

Airfield Pressure Height — The height registered at the surface of an aerodrome by an altimeter with the pressure sub-scale set to 1013 hPa.

Pressure Altitude — Altitude measured from standard sea-level pressure (1013 hPa) by an altimeter corrected for position and instrument error.

Wind — The wind velocities to be used as variables on aircraft performance are to be understood as the headwind or tailwind components of the reported winds.

1.4.4 Power Terminology

Take-Off Power — Maximum power permissible for take-off.
Maximum Continuous Power — Maximum power that is allowed to be used continuously during flight.
Maximum Climb Power — Maximum power permissible during climb.
Maximum Cruise Power — Maximum power permissible during cruise.

1.4.5 Engine Controls and Instruments

Throttle Lever — The lever which the pilot uses to control the engine power.
Mixture Control — The control that is used to vary the fuel/air ratio available to the engine.
Tachometer — The instrument that indicates the engine RPM.

1.4.6 Aircraft Performance and Flight Planning Terminology

Climb Gradient — The ratio of the change in height during a climb, to the horizontal distance travelled.
Maximum Crosswind Component — The maximum allowable crosswind component for take-off and landing.

1.4.7 Weight and Balance Terminology

Reference Datum — An imaginary vertical plane from which all horizontal distances are measured for balance purposes.
Station — A location along the aircraft fuselage given as a distance from the reference datum.
Arm — The horizontal distance from the reference datum to the centre of gravity (CoG) of an item.
Moment — The product of the weight of an item multiplied by its arm.
Centre of Gravity (CoG) — The point at which an aircraft would balance if suspended. The distance from the CoG to the reference datum can be found by dividing the total moment by the total weight of the aircraft.
CoG Arm — The arm obtained by adding the aircraft's individual moments and dividing the sum by the total weight.
CoG Limits — The extreme centre of gravity locations within which the aircraft must be operated at a given weight.
Useable Fuel — Fuel available that may be used in flight.
Unusable Fuel — Fuel remaining in the aircraft that cannot be safely used in flight.
Empty Weight — Weight of aircraft with unusable fuel and full oil.
Useful Load — Difference between take-off weight and empty weight.
Maximum Take-Off Weight — Maximum weight approved for take-off.
Maximum Landing Weight — Maximum weight approved for the landing.

1.5 Use of Metric/Imperial Units

This POH uses the metric system as the basic system of measurement. Where common usage or available instrumentation refer to the imperial system, both units are quoted.

Conversion	Value
1 Pound (lb)	= 0.45359 Kilogram (Kg)
1 Pound per sq in (psi)	= 6.8947 Kilopascal (kPa)
1 Inch (in)	= 25.4 Millimetres (mm)
1 Foot (ft)	= 0.3048 Metre (m)
1 Statute mile	= 1.609 Kilometres (km)
1 Nautical mile (nm)	= 1.852 Kilometres (km)
1 Millibar (mb)	= 1 Hectopascal (hPa)
1 Imperial gallon	= 4.546 Litres (l)
1 US gallon	= 3.785 Litres (l)
1 US quart	= 0.946 Litre (l)
1 Cubic foot (ft^3)	= 28.317 Litres (l)
1 Degree Fahrenheit (°F)	= [1.8 × °C] + 32
1 Inch Pound (in lb)	= 0.113 Newton Metres (Nm)
1 Foot Pound (ft lb)	= 1.356 Newton Metres (Nm)

SECTION 2 — LIMITATIONS

2.1 General

This section presents the various operating limitations, instrument markings, colour coding, and basic placards necessary for the safe operation of the aircraft, it's engine, standard systems and standard equipment. All limitations contained in this section have been approved by the New Zealand Civil Aviation Authority.

For specific operations, or for operations with equipment fitted that is covered by a supplement in Section 9 of the POH, limitations applicable will be found in the relevant supplement.

2.2 Airspeed Limitations

Speed	Semi-Acrobatic KIAS	Acrobatic KIAS	Remarks
Max Manoeuvring Speed (V_A)	123	140	Do not make full or abrupt control movements above this speed.
Max Flap Extended Speed (V_FE)	103	103	Do not exceed this speed with any flap extended.
Max Structural Cruising Speed (V_C)	141	161	Do not exceed this speed except in smooth air and then only with caution.
Never Exceed Speed (V_NE)	178	203	Do not exceed this speed in any operation.

2.3 Airspeed Indicator Markings

Marking	IAS Value or Range	Significance
White Arc	49–103	Full Flap Operating Range. Lower limit is max weight stalling speed in landing configuration. Upper limit is V_FE.
Green Arc	55–141	Normal Operating Range. Lower limit is max weight stalling speed with flaps up. Upper limit is V_C.
Yellow Arc	141–178	Operations must be conducted with caution and only in smooth air.
Red Line	178	Maximum speed for all operations (V_NE).

> **NOTE:** *These markings are for semi-acrobatic category. Refer to para 2.2 above, or placard for acrobatic category limitations.*

2.4 Engine

Engine Manufacturer: Textron (Avco) Lycoming
Engine Model: O-320-E1A or O-320-D1A

2.4.1 Engine Limitations

Power	RPM	MAN Press	Max Cyl Head Temp	Max Oil Temp	Fuel Min	Fuel Max	Oil Min	Oil Max
Maximum Take-Off	2700	Full Throttle	260°C (500°F)	118°C (245°F)	0.5 psi	5 psi	60 psi	90 psi
Maximum Continuous	2700	Full Throttle	232°C (450°F)	118°C (245°F)	0.5 psi	5 psi	60 psi	90 psi

Notes:

1. Minimum oil pressure at idle: 25 psi (hot oil)
2. Max oil pressure after start: 100 psi (cold oil)
3. Minimum oil temperature for continuous operation: 60°C (140°F)

2.4.2 Fuel Grade

Engine	Approved Fuel
O-320-E1A	Avgas 80/87, 100LL or 100/130
O-320-D1A	Avgas 100LL or 100/130

2.4.3 Lubricating Oil

Specification: Textron Lycoming Specification № 301-F and Lycoming Service Instruction № 1014 approves lubricating oils conforming to specifications MIL-L-6082 for straight mineral oil and MIL-L-22851 for ashless dispersant oil.

Straight mineral oil must be used during the first 50 hours of operation or until the oil consumption has stabilised. After the first 50 hours a change to an approved additive oil should be made. Refer to Lycoming Service Instruction 1014 for further details.

Viscosity Grade:

Average Ambient Temperature	Mineral Grades	Ashless Dispersant Grades
All Temperatures	—	SAE 15W50 or 20W50
Above 27°C	SAE 60	SAE 60
Above 16°C	SAE 50	SAE 40 or SAE 50
-1°C to 32°C	SAE 40	SAE 40
-18°C to 21°C	SAE 30	SAE 40, 30, 20W4
Below -12°C	SAE 20	SAE 30, 20W30

SAE:	20	30	40	50	60
Commonly Used Designation:	55	65	80	100	120

Sump Capacity: 8 US quarts (7.6 litres) total; 6 US quarts (5.7 litres) useable.

2.4.4 Propeller

Type: Constant speed, metal, two blade
Manufacturer: Hartzell

Type	Diameter
HC-C2YL-1B/7663-4.0	72.0 inches Maximum
	72.0 inches Minimum
	NOTE: No further reduction permitted.

2.5 Engine Instrument Markings

Instrument	Yellow Arc (Caution Range)	Green Arc (Normal Operating)	Red Line (Maximum Limit)
Tachometer	—	500–2700 RPM	2700 RPM
Oil Pressure	25–60 psi	60–85 psi	100 psi

Instrument	Yellow Arc (Caution Range)	Green Arc (Normal Operating)	Red Line (Maximum Limit)
Oil Temperature	5–50°C	50–118°C	118°C
Fuel Pressure	0.5–2 psi	2–5 psi	5 psi

2.6 Weight Limits

Limit	Value
Maximum Take-Off Weight (Semi-Acrobatic Cat.)	862 kg (1900 lb)
Maximum Acrobatic Weight	816 kg (1800 lb)
Maximum Landing Weight	862 kg (1900 lb)

2.7 Centre of Gravity Limits

Forward Limit: 2011.7 mm (79.20 inches) aft of datum at 635 kg (1400 lb) and below; 2095.5 mm (82.50 inches) aft of datum at 862 kg (1900 lb), with linear variation between these points.

Aft Limit: 2186.9 mm (86.1 inches) aft of datum at all weights.

Datum: Fuselage Station 0 *[1270.0 mm (50 inches) forward of engine fire wall]*

2.8 Manoeuvre Limits

The following acrobatic manoeuvres (and any combination of them) are approved in the semi-acrobatic or acrobatic categories. In all cases the flaps must be in the fully UP position. No luggage or loose objects may be carried in the luggage compartment.

- Spins *(Flaps UP only and limited to a maximum of 1½ turns)*
- Inside loops
- Slow roll
- Aileron roll
- Barrel roll

- Stall turn
- Half inside loop and roll out
- Half roll and pull through
- Chandelle & Lazy eight
- Inverted flight

> **NOTE:** *Extended inverted flight is not permitted due to engine fuel and oil limitations.*

2.9 Allowable Flight Load Factor Limits

Category	Flap UP	Flap DOWN
Semi-Acrobatic category — 1900 lb	+4.4g / -2.2g	+2.2g / -0g
Acrobatic category — 1800 lb	+6.0g / -3.0g	+3.0g / -0g

> **NOTE:** *The above loads are the maximum allowable in flight. The aircraft structure has been designed to withstand 150% of these loads without catastrophic failure.*

2.10 Cross Wind Component Limit

The maximum allowable cross wind component for take-off or landing is **20 knots**.

2.11 Kinds of Operations Limits

When fitted with original factory installed equipment this aircraft is approved for **VFR day operations only**.

If additional instrumentation and radio navigation equipment is fitted as required by the appropriate regulations governing these types of operations, the aircraft is approved for VFR Night or IFR operations.

2.12 Fuel Capacity

Total Fuel	131.8 litres
Useable Fuel	130.5 litres

2.13 Flight Crew Limits

Minimum flight crew is one pilot. Solo flight from left hand seat only (unless electric flaps are fitted).

2.14 Maximum Passenger Seating Limits

The maximum passenger seating capacity is one, seated beside the pilot.

2.15 Other Limitations

- The sliding canopy may be opened in flight.
- Maximum allowable airspeed with canopy open is **70 KIAS**.
- No baggage may be carried during acrobatics.

2.16 Placards

The following placards must be prominently displayed:

1. In front and in clear view of the pilot:

(a) "This aircraft must be operated as a semi-acrobatic or acrobatic category aircraft in compliance with the operating limitations stated in the form of placards, markings and manuals"

(b) "Airspeed indicator markings are for semi-acrobatic category. Refer to flight manual or placard for acrobatic limitations"

(c) "This aircraft is approved for VFR day and night operations only" *(If the required equipment is installed IFR operations may be added to this placard.)*

(d) "This aircraft is not approved for flight in airframe or propeller icing conditions"

2. Close to the airspeed indicator:

Airspeed Limitations — Knots IAS	Semi-Acrobatic 1900 lb	Acrobatic 1800 lb
Never Exceed V_NE	178	203
Maximum Structural Cruise V_C	141	161
Manoeuvring Speed V_A	123	140
Maximum Flaps Extended V_F	103	103
Maximum Crosswind Landing or Take-off	20	20

3. In clear view of the pilot:

(a) "Smoking is permitted except during refuelling, take-off, landing and acrobatics"

(b) "Solo from LH seat only" *(May be omitted if electrically operated flaps are installed.)*

(c) "Spinning permitted at all weights up to one and a half turns only. Use normal recovery techniques. Intentional spins with flaps extended are prohibited"

(d) Approved manoeuvres and recommended entry speeds (KIAS):

Manoeuvre	Speed (KIAS)
Barrel roll	115
Slow roll	90
Aileron roll	100
Flick roll	80
Half roll and pull through	85
Half inside loop and roll out	140
Loop	130
Stall turn	130
Steep turn 1800 lb and below	140
Steep turn over 1800 lb	120
Inverted flight	100
Chandelle and lazy eight	110

4. Near the carburettor heat control:

- "Take-off and climb: Full cold recommended"
- "Other conditions: Heated air as required particularly during approach and landing"

5. In the baggage compartment:

- "Baggage — 100 lb maximum"
- "Baggage must not be carried when performing aerobatics"

SECTION 3 — EMERGENCY PROCEDURES

3.1 General

Section 3 of this Flight Manual describes the procedures to be adopted in the event of an emergency or abnormal situation occurring in the Airtourer T6 aircraft.

The procedures are arranged in the sequence considered to be the most desirable in the majority of cases. Steps should be performed in the order listed unless good reasons for deviation exist.

It should be remembered however, that all conceivable eventualities cannot be foreseen by the manufacturer. Particular circumstances such as multiple or unanticipated emergencies, adverse weather etc. may require modification to these procedures. A thorough knowledge of the aircraft and it's systems is essential to analyse the situation correctly and determine the best course of action in any particular circumstance.

The following basic rules apply to all aircraft emergencies:

1. Maintain Aircraft Control.
2. Analyse the situation and take appropriate action.
3. Land as soon as practicable.

3.2 Emergency Procedures Check Lists

3.2.1 Engine Failure During Take-off Run

Action	Setting
Throttle	IDLE
Brakes	APPLY

Action	Setting
If overrun appears likely and time permits:	
Wing Flaps	UP
Fuel Shut-off	OFF
Ignition	OFF
Master Switch	OFF

3.2.2 Engine Failure Immediately After Take-Off

Action	Setting
Lower the nose — Airspeed	70 KIAS — *Immediate action!!*
Landing area	SELECT BEST AVAILABLE STRAIGHT AHEAD

> **WARNING:** *Land straight ahead — avoid major obstacles — do not turn back!*

If time permits conduct a trouble check ascertaining switches ON, mixture RICH, fuel ON

Before Landing:

Action	Setting
Switches	OFF
Fuel	OFF
Braking	HEAVY AFTER TOUCHDOWN

3.2.3 Engine Failure in Flight — Restart

Action	Setting
Airspeed	85 KIAS
Fuel	ON
Fuel Pump	ON
Mixture	RICH
Switches	Both magnetos and Master ON

Action	Setting
Throttle	SET ½" Open
Pitch	FULL FINE

Engage starter (if propeller is stopped) and release immediately when engine fires.

Notes: (a) *Re-establish cruise power and RPM, then lean mixture as required.* (b) *As carburettor icing may have been the contributing factor to engine stopping, use carburettor heat as necessary to avoid any further icing.* (c) *If engine does not restart, commence forced landing procedure.* (d) *If clear symptoms of a mechanical failure exist, or if the engine has seized due to the loss of oil pressure, do not attempt a restart.*

3.2.4 Engine Failure — Forced Landings

Action	Setting
Carby heat	HOT immediately
Airspeed	85 KIAS & trim
Trouble checks	Switches ON, Mixture fully RICH, fuel ON

Once it has been decided that there is no likelihood of an engine re-start and a forced landing is imminent.

Action	Setting
Proposed landing area	SELECT and proceed with forced landing procedure

Safety Checks:

Letter	Item	Action
H	Hatches & Harnesses	Closed and Tight
T	Throttle	Closed
F	Fuel	OFF
S	Switches	OFF
B	Brakes	OFF

When on the base leg employ the glide approach technique. If an excessive overshooting tendency is apparent, correction may be assisted by executing a slipping turn on to the final approach. When certain that the selected landing area can be reached lower the flaps fully.

Notes:

1. *Side slipping with the flaps down* ***is permitted****.*
2. *In simulating forced landings the engine should be cleared at least every 1000 feet during the descent.*
3. *Selecting propeller pitch to* ***coarse*** *will reduce drag and improve glide performance if required.*

3.2.5 Precautionary Landing with Engine Power

Action	Setting
Airspeed	70 KIAS
Wing Flaps	8°
Selected field	FLY OVER & INSPECT
Landing	SHORT FIELD TECHNIQUE

3.2.6 Ditching

Action	Setting
Airspeed	70 KIAS
Wing Flaps	8°
Power (if available)	ESTABLISH 300 ft/min @ 70 KIAS
Approach — High Winds, Heavy Seas	INTO WIND
Approach — Light Winds, Heavy Swells	PARALLEL TO SWELLS
Flaps	FULL PRIOR TO TOUCH DOWN
Touch Down	SLOWEST PRACTICAL SPEED

> **WARNING:** *This is a recommended procedure based on the best available judgement. No ditching trials or tests have been carried out.*

3.2.7 Engine Fire During Start on Ground

Action	Setting
Throttle	CLOSED
Mixture	IDLE CUT-OFF
Switches	OFF
Fuel	OFF
Aircraft	EVACUATE and extinguish fire using best available means

If necessary, release the brakes to enable the aircraft to be pushed away from burning fuel on the ground.

Notes:

1. *Sealing off the carburettor air intake with a piece of material will effectively extinguish an intake fire if a chemical extinguisher is not available.*
2. *On no account should a re-start be attempted until the engine has been examined and certified as serviceable by a licensed aircraft engineer.*

3.2.8 Engine Fire in Flight

Action	Setting
Throttle	CLOSED
Fuel	OFF
Mixture	ICO
Switches	OFF
Airspeed	REDUCE
Forced Landing	EXECUTE *(if the fire persists sideslip as required to minimise the effect of the fire)*

3.2.9 Smoke/Fume Evacuation

Action	Setting
Power	REDUCE

Action	Setting
Airspeed	APPROX 70 KIAS
Canopy	OPEN
Power	ADJUST to maintain approx 70 KIAS
Canopy	CLOSE WHEN CLEAR

> **NOTE:** *Aircraft may be landed with canopy open if necessary.*

SECTION 4 — NORMAL PROCEDURES

4.1 General

Section 4 of this Flight Manual describes the procedures to be adopted for normal operations of the Airtourer T6 aircraft. The procedures are arranged in the sequence considered to be the most desirable and therefore steps should be performed in the order listed unless good reasons for a deviation exist.

4.2 Summary of Speeds for Normal Operation

(Based on a maximum weight of 862 kg)

Take-Off:

- TOSS (Speed @ 50 ft): **70 KIAS**

Climb, Flaps Up:

- Vy (Best Rate, Sea Level–5000 ft): **80 KIAS**
- Vy (Best Rate, Above 5000 ft): **75 KIAS**

Climb, Flaps 8°:

- Vx (Best Angle, Sea Level): **65 KIAS**

Landing Approach:

- V_APP (Speed @ 50 ft): **70 KIAS**
- Go-Around / Baulked Landing: **65 KIAS Initially**

Best Glide Angle:

- Flaps UP: **85 KIAS**

4.3 Normal Procedures Checklists

4.3.1 Preflight Inspection

Before flight, a careful preflight inspection should be carried out to ensure that the aircraft is serviceable.

1. Cockpit

Item	Action
Magnetos	OFF
Master	ON
Fuel	ON
Nav/Landing Lights	ON/OFF as req to check
Mixture	ICO
Propeller	FULL FINE
Harnesses & Seats	CHECK CONDITION
Windshield	CLEANLINESS
Cockpit area	GENERAL CONDITION
Loose objects	SECURE
Canopy/Latches	CONDITION & OPERATION
Master	OFF after lights & stall warn checked

2. Left Wing / Left Centre Fuselage

Item	Action
Fuel Tank Sump Drain	DRAIN (water, sediment & fuel grade)
Flapperon & Aileron	CONDITION, HINGES & SECURITY
Wingtip	CONDITION & NAV LIGHTS
Leading edge	DAMAGE & CONDITION
Stall Warning Vane	OPERATION (horn should be heard)
Undercarriage	SECURITY & OBVIOUS DEFECTS
Tyres	INFLATION & CONDITION
Brake Assy	CONDITION, PADS, CALIPER & HOSE

3. Engine/Propeller/Nose

Item	Action
Engine Air Outlets/Inlets	CHECK FOR OBSTRUCTIONS
Exhaust System	CONDITION & SECURITY
Accessible Engine Components	CONDITION & SECURITY
Propeller	NICKS & DAMAGE, BLADE SECURITY
Spinner	CONDITION & SECURITY
Cowling	CONDITION & SECURITY
Oil	QUANTITY/DIPSTICK SECURE *(not over tight)*

4. Right Wing / Right Centre Fuselage

Item	Action
Undercarriage	SECURITY & OBVIOUS DEFECTS
Tyres	INFLATION & CONDITION
Brake Assy	CONDITION, PADS, CALIPER & HOSE
Leading edge	DAMAGE & CONDITION
Fuel Vent	CHECKED & CLEAR
Fuel Cap	SECURE
Fuel Contents	CHECK WITH DIPSTICK AS REQUIRED
Wingtip	CONDITION & NAV LIGHTS
Flapperon & Aileron	CONDITION, HINGES & SECURITY

5. Empennage

Item	Action
Tailplane	CONDITION
Elevators	CONDITION & SECURITY
Rudder	CONDITION & SECURITY

4.3.2 Before Starting Engine

Item	Action
Pre-flight Inspection	COMPLETED
Passenger Briefing	COMPLETED
Rudder Pedals	ADJUSTED
Baggage Compartment	CONTENTS SECURE
Harnesses	SECURE
Brakes	ON/PARK

Item	Action
Carb Heat	COLD
Flaps	UP
Radios & Navaids	OFF
Throttle	FREE & FRICTION SET
Propeller	FULL FINE
Fuel	ON
Fuel Contents Gauge	CHECK

4.3.3 Starting Engine

Item	Action
Park Brake	CHECK ON
Master	ON
Fuel Pump	ON
Anti-Collision Light	ON
Magnetos	LEFT *(T6 only, not in T6/24 variant)*
Mixture	FULL RICH
Throttle (Cold Start)	2 FULL STROKES, then OPEN ½"
Clear	ALL ROUND — "CLEAR PROP"
Starter	ACTIVATE until engine fires
Magnetos	BOTH *(T6 only, not in T6/24 variant)*

> **NOTE:** *Hot Start — Initial procedure is similar to cold start but do not prime the engine with throttle. If the engine does not fire during the first few revolutions give one pump on the throttle before trying again.*

4.3.4 After Start

Item	Action
Throttle	SET **800 RPM** for the first minute
Oil Pressure	RISING within 30 seconds — *if not* SHUT DOWN
Generator	ON — light out
Fuel Pump	OFF — pressure checked
Radios & Navaids	ON and tuned
DG	SET to compass heading
AH	SET 0° pitch — wings level
Altimeter	AIRFIELD ELEVATION or QNH SET
Throttle	SET **1200 RPM** once oil temp reaches 60°C

4.3.5 Engine Run-Up

Item	Action
Park Brake	ON
Temps & Pressures	WITHIN LIMITS
Throttle	2000 RPM
Park Brake	HOLDING — no creep forward
Carb Heat	ON (10 secs) — RPM drop, then OFF; RPM returns to 1800 RPM
Magnetos	CHECK (Both-Left Off-Both-Right Off-Both); Max drop 125 RPM
Propeller	EXERCISE twice through full range to circulate oil
Generator	CHARGING — LIGHT OUT
Throttle	CHECK IDLE — 650 to 850 RPM

4.3.6 Pre Take-Off Vital Actions (T-M-P-F-I-S-C-H)

Letter	Item	Action
T	Trim / Throttle Friction	NEUTRAL for take-off / CONFIRMED SET
M	Mixture	FULL RICH
P	Propeller	FULL FINE
F	Fuel / Flaps	ON, contents SUFFICIENT, pump ON, pressure OK / 8° — visually checked
I	Instruments	QNH set & checked, AH set, DG set
S	Switches	Master ON, generator ON, fuel pump ON, lights AS REQ, magnetos BOTH, circuit breakers IN
C	Controls / Carby Heat	FULL & FREE MOVEMENT, CORRECT SENSE / COLD
H	Hatches & Harnesses	SECURE

4.3.7 Line Up

Item	Action
Transponder	ALT
DG	ALIGNED to runway heading
Compass	CHECKED within 10° of runway heading

4.3.8 Take-Off

Item	Action
Throttle	FULL OPEN
Elevator Control	EASE WEIGHT OFF NOSE WHEEL @ 45–50 KIAS

Item	Action
Take Off Safety Speed	ATTAIN (70 KIAS)

4.3.9 After Take-Off

Item	Action
Flaps	UP *(above 200 feet AGL with positive Rate of Climb)*
Initial Climb Speed	70 KIAS
Fuel Pump	OFF — pressure checked

4.3.10 Climb

Item	Action
Power	25"Hg — Full Throttle / 2500–2700 RPM
Airspeed	80 KIAS *(reducing to 75 KIAS above 5000 ft)*

4.3.11 Cruise

Item	Action
Power	24"Hg/2400 RPM *(refer Section 5, para 5.7)*
Mixture	RICH/LEAN *(as required — refer para 4.4.12)*

4.3.12 Descent

Item	Action
Power	REDUCE *(not below 15"Hg)*
Mixture	ENRICH during descent

4.3.13 Before Landing (Down Wind Checks — B-U-M-F-H-T)

Letter	Item	Action
B	Brakes	OFF (Park Brake) — brake pressure available
U	Undercarriage	FIXED
M	Mixture	FULL RICH
F	Fuel	ON — contents SUFFICIENT, pump ON, pressure CHECKED
H	Hatches & Harnesses	SECURE
T	Threshold Speed	70 KIAS *(Consider increase for weather)*

4.3.14 Finals Check (approx 200 ft AGL — P-U-F)

Letter	Item	Action
P	Propeller	FULL FINE
U	Undercarriage	FIXED
F	Flaps / Carburettor Heat	LANDING SETTING (normally 12–15°) / COLD

4.3.15 Go-Around / Baulked Landing

Item	Action
Power	THROTTLE FULL OPEN
Airspeed	65 KIAS Initially until positive rate of climb
Flaps	UP — Slowly during acceleration to normal climb speed
Fuel Pump	OFF — pressure checked

4.3.16 After Landing (Clearing the Runway)

Item	Action
Flaps	IDENTIFIED and UP
Transponder	STBY
Fuel Pump	OFF
Trims	NEUTRAL

4.3.17 Shut Down (Once Parked)

Item	Action
Park Brake	ON
Throttle	800–900 RPM
Radios & Navaids	OFF
Magnetos	CHECK for drop, dead cut, back to BOTH
Mixture	IDLE CUT OFF
After engine stops:	
Magnetos	OFF
Master	OFF
Switches	ALL OFF

4.4 Amplified Procedures

4.4.1 Taxying

The wide track tricycle undercarriage, coupled with the nose wheel steering and efficient braking system allows high wind conditions to be handled easily. The aircraft has been demonstrated as safe to taxi in winds up to 30 knots. When taxying in strong winds additional care should be taken, particularly in quartering tail wind conditions. Appropriate

elevator and aileron control positions are essential when taxying in strong tail wind conditions.

Taxying over loose gravel or stones should be done at low engine RPM to minimise propeller damage.

4.4.2 Run-Up

Full throttle run-ups over loose gravel are especially harmful to the propeller and should be avoided. When take-offs must be made from a gravel surface, it is very important that the throttle be advanced slowly and a rolling start take-off technique be used to minimise propeller damage.

Prior to take-off from high elevation airfields, the mixture should be leaned just enough to ensure smooth engine operation in a full throttle static runup.

4.4.3 Normal Take-Off

Select 8° of flap, align the aircraft carefully with the take-off path, release the brakes and then open the throttle slowly to the fully open position.

> **NOTE:** *It is important to check full throttle engine performance early in the take-off run. Any sign of rough engine operation or sluggish acceleration is good cause for discontinuing the take off and conducting a full power runup to confirm normal engine operation prior to the next take-off attempt.*

Keep straight by use of the rudder. There is a slight tendency to swing to the left if the throttle is opened too quickly. Weight should be eased off the nose-wheel at 45–50 KIAS.

The aircraft should be flown off the ground at approximately 60 KIAS. Allow the airspeed to build up to 70 KIAS before climbing. At approximately 200 feet flaps may be raised, and airspeed increased to the recommended climb speed of 80 KIAS.

4.4.4 Short Field Take-Off

A short field take-off technique **must be used in order to achieve the take-off distances specified in Section 5.** The 8° of flap must be used. The shortest take-off run will be obtained if the aircraft is held against the brakes until full power has been applied.

If obstacle clearance is a consideration a speed of 65 KIAS (which gives the best angle of climb) should be maintained until clear of all obstacles. Speed should then be increased to recommended climb speed of 80 KIAS after raising flap.

> **CAUTION:** *Excessive rotation on take-off may cause the rear tail skid to scrape the ground. Normally any scraping which may occur is of a minor nature but if a severe scrape occurs, the aircraft should be inspected and certified as airworthy before further flight.*

4.4.5 Cross Wind Take-Off

Nosewheel steering can be used effectively during crosswind take-offs and it is advantageous to keep some weight on the nosewheel during initial acceleration until rudder control is fully obtained. The following procedure should be applied:

1. Open the throttle slowly, keeping straight with combined use of rudder and nosewheel steering and by holding the control column into wind.
2. Hold the aircraft on the ground until a speed of 55 KIAS is reached, then unstick with a positive movement.
3. Adjust the airspeed to 70 KIAS (with flap) laying off the effect of drift by yawing the nose into wind as required.

> **NOTE:** *The maximum allowable crosswind component is 20 knots.*

4.4.6 Climbing

The speed for maximum rate of climb is 80 KIAS flaps up. However a climb speed of 85 KIAS provides more positive control with a lower nose attitude, better visibility and better engine cooling, and may be used when maximum climb performance is not required.

Climbing may be safely accomplished at full throttle and 2700 RPM. Nevertheless a lower power setting may be used if desired, usually not below 25"Hg/2500 RPM. Rich mixture should always be used for climbing unless leaning is necessary to eliminate rough running due to an over richness at altitude, normally occurring not below 5000 feet.

4.4.7 General Flying

Flying Controls — The controls are well harmonised and the aircraft is easy and pleasant to fly. The controls remain light and effective throughout the speed range, except that the ailerons tend to become heavier as V_NE is approached.

Changes of Trim:

- Flaps Down: Slightly Nose Down
- Flaps Up: Slightly Nose Up

Changes of power and speed promote very slight changes in directional trim.

Stability — The aircraft is easy to trim under all conditions of flight, and holds its trimmed speed well.

4.4.8 Flying at Reduced Airspeeds

Reduce speed to 70 KIAS and lower 8° of flap, then adjust power to maintain 65–70 KIAS. Stalling speed under these conditions is approximately 53 KIAS.

4.4.9 Carburettor Icing

Carburettor icing is indicated by rough running and/or loss of power. To remedy, select HOT air with the carburettor heat control until icing has cleared, then re-select COLD air. If icing does not clear shortly after the selection of HOT, manipulation of the throttle may assist.

Do not use partial heat. Use FULL HOT or FULL COLD only.

> **CAUTION:** *Carburettor heat will increase fuel consumption and reduce maximum available engine power.*

Avoid flying close to or underneath well developed cumulus/cumulonimbus clouds.

4.4.10 Mixture Control

The mixture control should be set to fully rich for starting, ground running, take-off, climb and landing. Changes in altitude will require corresponding adjustments of the mixture control, normally not below 5000 feet. It should always be returned to the fully rich position before commencing a dive and/or acrobatics.

The engine should never be allowed to run for any length of time at reduced power where the reduction in power is obtained by use of the mixture control. An over weak mixture will lead to the engine over-heating, detonation and engine damage.

4.4.11 Vapour Lock

When operating in very hot weather there is a possibility of vapour forming in the fuel suction lines. Idling on the ground should, therefore, be reduced to the minimum, and RPM maintained at a higher figure than that normally recommended. For the same reason flight above 10,000 feet is not recommended for extended periods. Power loss due to vapour formation may be restored by turning the electric fuel pump on, and/or descending to a lower altitude.

4.4.12 Normal Cruising

Maximum Cruise Speed — The maximum cruise speed is 125 KIAS at Full throttle/2700 RPM. The mixture should be full rich when cruising at full power.

Normal Cruise — A normal cruise power setting of 24"Hg/2400 RPM will give a cruise speed of approximately 120 KIAS. Below 5000 ft the mixture may be left full rich, but above 5000 ft the mixture should always be correctly leaned.

Leaning Procedure:

1. Lean the mixture by slowly pulling the mixture control back towards the lean position until slight rough running occurs.
2. Slowly move the mixture control back towards rich until smooth running is just restored.

Flying for Range — The airspeed required to achieve maximum range is 110 KIAS:

1. Reduce RPM to 2300 and adjust the throttle to obtain the recommended speed. The manifold pressure will be approximately 23"Hg.
2. Lean the mixture by slowly pulling the mixture control back towards the lean position until slight rough running occurs.
3. Slowly move the mixture control back towards rich until smooth running is just restored.

Flying for Endurance — The airspeed required to achieve maximum endurance is 95 KIAS:

1. Reduce RPM to 2100 and adjust the throttle to obtain the recommended speed.
2. Lean the mixture by slowly pulling the mixture control back towards the lean position until slight rough running occurs.
3. Slowly move the mixture control back towards rich until smooth running is just restored.

4.4.13 Gliding

The best gliding angle (greatest gliding distance for any given height loss) will be achieved with flaps UP at 85 KIAS. The gliding angle will be improved slightly with the propeller pitch set to COARSE. The gliding angle with flaps extended will be slightly worse than with the flaps UP.

4.4.14 Side Slipping

Side slipping may be used to increase drag such as may be desirable during a forced landing approach. Side slips may be carried out with flaps extended.

4.4.15 Pre-Stalling & Acrobatic Checks (H-A-S-E-L-L)

Letter	Item	Requirement
H	Height	Sufficient to recover by 3,000 feet
A	Airframe	The aircraft is configured — ie weight, canopy, flaps etc.
S	Security	Hatches, harnesses, no loose articles
E	Engine	Temperatures, pressures, carburettor heat, throttle friction etc. / Propeller set 2650 RPM
L	Location	Clear of built up areas, location of forced landing area
L	Lookout	Clear of all other aircraft and cloud both vertically & horizontally

4.4.16 Stalling

Before practising stalling complete the Pre-Stalling and Acrobatics checks.

Approximate stalling speeds at maximum AUW @ 862 kg:

Configuration	Stall Speed
Power Off, flaps up	55 KIAS
Power Off, flaps full down	49 KIAS
Power On, under typical approach conditions with full flap	45 KIAS

Stall warning in all cases consists of aural warning 5–10 knots before the stall occurs. There is no obvious pre-stall airframe buffet.

With power off at the stall, the nose drops gently, with a slight tendency for the nose to oscillate in pitch. With power on, there is a well defined tendency for a wing to drop. Stalling speeds are reduced by about two to three knots with one pilot only in the aircraft. Recovery in all cases is straight forward and easy using standard techniques. Maximum height loss in normal recovery from power off stall is 200 ft.

> **CAUTION:** *The stall warning horn is disabled when the flaps are UP.*

4.4.17 Spinning

One and a half turns only are permitted with aircraft in fully acrobatic configuration, ie. maximum 816 kg AUW.

Recommended Entry Technique — With the aircraft trimmed, power off, flaps up, gradually raise the nose above the horizon with wings level. At the point of stall, apply full rudder in direction of the desired spin, at the same time holding the control column centrally hard back. The aircraft will perform an unstable spin with the associated buffet and speed increase. Rate of rotation will be relatively high with a steep nose down attitude. Some aircraft may tend to recover on their own after one full turn.

Recovery — Use standard recovery technique. Recovery is almost immediate.

> **CAUTION:** *Excessive 'g' loadings and airspeed build up can occur if recovery is delayed beyond two turns.*

Inadvertent Spin — In the event of an inadvertent spin, ensure flaps up, throttle closed. Apply standard recovery technique. Recovery is almost immediate.

4.4.18 Diving

Before commencing a dive, the mixture control should be in the fully rich position. With the

increase in speed the aircraft can be easily held in the dive to the limiting speed without re-trimming. Directional trim can be maintained by progressive application of rudder.

4.4.19 Acrobatics

All normal acrobatic manoeuvres may be performed up to the maximum all up weight of 816 kg. Recommended entry speeds:

Manoeuvre	Speed
Roll	100 KIAS
Barrel Roll	110 KIAS
Stall Turn	100 KIAS
Loop	125 KIAS
Half Roll Off Top of Loop	140 KIAS
Half Roll & Pull Through	70 KIAS
Flick Roll	70 KIAS

> **NOTE:** *Owing to possible loss of oil during acrobatics, the engine oil level should be checked as full prior to any flight involving acrobatics.*

> **CAUTION:** *Loss of engine oil pressure during zero and negative 'g' manoeuvres may cause an engine overspeed due to loss of oil pressure to the propeller. If overspeed occurs reduce power (with throttle) immediately, note RPM reached, and have the aircraft inspected by a licensed aircraft engineer before further flight.*

4.4.20 Approach & Landing

Glide Approach — On the base leg close the throttle and reduce the speed to 70 KIAS and lower flaps (as required). Maintain 70 KIAS throughout the approach, aiming to cross the threshold at 70 KIAS.

Powered Approach — Reduce power and airspeed to approximately 1400 RPM and 70 KIAS and select flap as required. Adjust airspeed to 70 KIAS aiming to cross the threshold at 70 KIAS.

> **CAUTION:** *For all approaches select carburettor heat to HOT prior to reducing power.*

Short Field Landing — Proceed as for a powered approach, reducing speed to 70 KIAS by the 50 foot height on late final with power off.

> **CAUTION:** *Care must be taken to ensure airspeed is maintained during the landing approach as the aircraft has high drag at approach speed and has a marked tendency to*

> *lose airspeed below 70 KIAS.*

Flapless Landing — Proceed as for a powered approach but maintain 75 KIAS throughout the approach.

> **CAUTION:** *During both short field and flapless landings care must be taken to avoid the rear tail skid from scraping the ground during the landing flare.*

Cross Wind Landing — Initially as for a normal crosswind approach and landing. When landing in a strong cross wind use a wing low, crab, or a combination method of drift correction. The aircraft is easy to handle in all cross wind conditions up to the maximum approved limit.

4.4.21 Going Around/Baulked Landing

Open the throttle fully and smoothly. Ensure carburettor heat has been selected to COLD. At full throttle the aircraft will climb away easily with the flaps fully down. Re-trim as necessary. Initially climb at 65 KIAS with the flaps lowered. At a safe height of around 200 feet, raise the flaps in easy stages. There is little change in trim and no sink. Provided obstacle clearance is not critical allow the airspeed to build up to a climb speed of 85 KIAS when the flaps are fully up.

SECTION 5 — PERFORMANCE

5.1 General

The performance data in this section was derived from actual flight tests. An aircraft that was in good condition and had an engine that was capable of developing it's rated power was used. Average, but accurate, piloting techniques were used during flight tests.

5.2 Take-Off and Landing Performance

5.2.1 Take-Off Distance

Un-factored take-off distances (metres) to clear a 50 ft obstacle. Based on: max take-off weight 1900 lb (862 kg), level hard surface, nil wind, 8° flap, 2700 RPM and full throttle before brake release. Aircraft lifted off at 60 KIAS, accelerated to 70 KIAS and climbed at this speed to 50 ft prior to raising flap.

Temp/Altitude	ISA	ISA+5°C	ISA+10°C	ISA+15°C	ISA+20°C
Sea Level	421	440	460	475	512
1000 ft	454	477	500	527	556
2000 ft	495	518	543	573	603
3000 ft	543	570	596	629	664

Corrections for non-standard conditions:

Weight — For every 25 kg for which the take-off weight is less than the maximum of 862 kg, subtract 12 metres from the required take-off distance.

Wind — For every one knot of head wind component (up to 8 knots) subtract 6 metres from the required take-off distance as corrected for weight.

Grass Surface — For operation from short dry grass add 5% to the take-off distance as corrected for weight and wind.

5.2.2 Landing Distance

Un-factored landing distances (metres) from a screen height of 50 ft to stopped. Based on: max landing weight 1900 lb (862 kg), level hard surface, nil wind, 15° (Full) flap, approach speed of 70 KIAS at 50 ft, heavy braking after touch down until stopped.

Temp/Altitude	ISA	ISA+5°C	ISA+10°C	ISA+15°C	ISA+20°C
Sea Level	346	352	358	366	372
1000 ft	357	363	369	376	384
2000 ft	367	373	381	387	395
3000 ft	379	387	393	402	410

Corrections for non-standard conditions:

Weight — Because the approach speed is constant and does not vary with weight, there is no significant change in landing distance.

Wind — For every one knot of head wind component (up to 8 knots) subtract 8 metres from the required landing distance.

Grass Surface — For operation from short dry grass add 10% to the landing distance as corrected for wind.

5.3 Stalling Speeds

At 862 kg (1900 lb)

Flap Setting	Stall Speed
UP	55 KIAS
4°	54 KIAS
8°	52 KIAS
12°	51 KIAS
15°	49 KIAS

Notes:

1. These speeds assume CoG on the forward limit, level flight, zero bank angle, and deceleration at not more than 1 knot/second.
2. CoG position aft of the forward limit, and/or deceleration greater than 1 knot/second will reduce the stalling speed.
3. Increasing angle of bank will increase the stalling speed. At 60° angle of bank the stalling speed will be increased by 41%.

5.4 Stall Warning

There is no obvious pre-stall airframe buffet. Stall warning consists of an aural warning 5–10 knots before the stall occurs.

> **CAUTION:** *The stall warning horn is disabled when the flaps are UP.*

5.5 Position Error Correction

The following table provides position error correction data to convert Indicated Airspeed (IAS) to Calibrated Airspeed (CAS). To obtain CAS, **add** the corrections to the IAS.

Flap \ KIAS	60-69	70-79	80-89	90-109	110-119	120-129	130-139	140-150
Clean	+3	+3	+3	+2	+1	0	-2	-4
8°	+3	+2	+1	—	—	—	—	—
15°	+2	+1.5	+1	—	—	—	—	—

5.6 Cruise Performance

	Speed
Normal Cruise Speed	120 KIAS
Maximum Range Speed	110 KIAS
Maximum Endurance Speed	95 KIAS

5.7 Fuel Consumption

Operation	Man Press	RPM	Mixture	Airspeed	Fuel Flow
Climb	FT	2700	Rich	80 KIAS	53.4 lt/hr
Normal Cruise	24"Hg	2400	Rich	120 KIAS	35.9 lt/hr
Normal Cruise	24"Hg	2400	Lean	120 KIAS	34.0 lt/hr
Maximum Range	23"Hg	2300	Lean	110 KIAS	32.3 lt/hr
Maximum Endurance	As Req	2100	Lean	95 KIAS	22.7 lt/hr

> **CAUTION:** *These figures are based on correct observance of all engine limitations and fuel leaning techniques. Variations may occur between individual aircraft and during the life of the engine.*

5.8 Best Angle of Climb (Vx)

Configuration for the best angle of climb performance: 8° Flap selected; 65 KIAS climb

speed; Full power.

SECTION 6 — WEIGHT & BALANCE

6.1 Empty Weight Data

The current empty weight data for this aircraft is contained in the LOAD DATA SHEET. This data sheet does not form part of this Flight Manual and is independently approved by a Weight Control Officer. It is recommended that the current LOAD DATA SHEET be inserted in the Flight Manual binder behind this Flight Manual.

> **NOTE:** *The empty weight data includes unuseable fuel and* ***full engine oil****.*

6.2 Loading Data

6.2.1 Moment Arms

Item	Arm
Pilot	2299 mm (90.5 inches) aft of datum
Passenger	2299 mm (90.5 inches) aft of datum
Fuel	2489 mm (98.0 inches) aft of datum
Baggage	2819 mm (111.0 inches) aft of datum

6.2.2 Fuel Weights

(Fuel specific gravity of 0.7)

Imp Gal	Litres	Kg	Litres	Kg
5	22.7	16	20	14
10	45.5	32	40	28
15	68.2	48	60	42
20	90.9	64	80	56
25	113.7	80	100	70
28.7	130.5	91	120	84
			130.5	91

6.3 Calculating Actual Weight and Balance

1. Obtain the Empty Weight data for the aircraft from the Load Data Sheet (LDS) and enter data in the table. *(Example: Empty Weight 612 kg, Arm 2054 mm, Moment 1256690 kg mm)*
2. Enter weights of occupants, fuel and baggage in the table.
3. Multiply weights by the corresponding arms to give the moments.
4. Total up the weights and moments. Divide total moment by total weight to give arm: `Arm = Total Moment ÷ Total Weight`
5. Plot Total weight (Take-Off Weight) vs Arm on graph.

The aircraft will be correctly loaded for acrobatic flight provided the plotted point is anywhere within the **acrobatic flight envelope**. The aircraft will be correctly loaded for semi-acrobatic flight provided the plotted point is anywhere within the **semi-acrobatic flight envelope**.

Aircraft Loading Chart — Sample

Item	Weight (kg)	Arm (mm)	Moment (kg mm)
Empty — from LDS	612	2054	1,256,690
Pilot	77	× 2299 =	177,023
Passenger	77	× 2299 =	177,023
Fuel	92	× 2489 =	228,988

Item	Weight (kg)	Arm (mm)	Moment (kg mm)
Baggage	0	× 2819 =	0
Take-Off Weight	**858**	**2144**	**1,839,724**

The sample loading falls inside the semi-acrobatic category envelope but outside the acrobatic envelope. Therefore the aircraft is correctly loaded for semi-acrobatic but not acrobatic flight.

SECTION 7 — AIRCRAFT & SYSTEMS DESCRIPTIONS

7.0 General

There are two variants of the Airtourer T6 model, the T6 and the T6/24. These variants are essentially identical except that the T6/24 has a 24/28 volt electrical system in lieu of the 12/14 volt system fitted to the basic T6 model. Additionally the T6/24 has electrically operated flaps and elevator trim system.

7.1 Main Services

7.1.1 Electrical System

T6 — Power is supplied by a 40 amp engine driven alternator with its output regulated to a nominal 14 volts. The power is distributed via a DC bus bar at the rear of the electrical panel. A red 'alternator warning light' on the electrical panel will indicate the master switch has been left ON after engine shut-down. The 12 volt, 24 amp hour lead acid battery is contained within a fibreglass box mounted on the left hand side of the rear fuselage, accessible by a hinged panel. The battery will provide ample power to maintain normal services in excess of half an hour after alternator failure. All circuits are protected by circuit breakers on the electrical panel.

T6/24 — Power is supplied by a 50 amp engine driven alternator with its output regulated to a nominal 28 volts. The 24 volt, 13 amp hour nickel-cadmium battery is contained within a fibreglass box mounted on the left hand side of the rear fuselage, accessible by a hinged panel.

7.1.2 Vacuum System

An engine driven vacuum pump provides suction for the Turn & Bank indicator, Artificial Horizon and the Directional Gyro. Suction is controlled by a regulator mounted on the rear face of the firewall behind the instrument panel.

7.2 Fuel System

7.2.1 Fuel Tank

A flexible bag type fuel cell is installed in the wing centre section between the front and rear spar. Total capacity is 131.8 litres, useable capacity is 130.5 litres.

7.2.2 Fuel Venting

The tank is vented to the atmosphere from the filling neck via a vent located on the starboard side of the rear fuselage.

7.2.3 Fuel Contents Indicating System

An electrically operated fuel contents indicating system provides a continuous indication of the fuel state on the fuel gauge mounted on the fuel panel.

7.2.4 Fuel Pumps

An engine driven fuel pump is provided on the engine while an auxiliary electric fuel pump is installed on the firewall. The latter may be selected ON from the cockpit.

7.2.5 Fuel Cock

A fuel cock lever marked ON when in fore and aft line of aircraft, and OFF when positioned at right angles to the fore and aft line is situated on the fuel panel adjacent to the base of the control column in the cockpit.

7.3 Engine Oil System

The oil sump on the engine has a capacity of 8 US quarts (7.6 litres). The oil level is checked by a dip stick embodied in the filler cap accessible by undoing the starboard side engine cowl. A dual oil pressure/temperature gauge is mounted on the instrument panel.

7.4 Engine Controls

7.4.1 Throttle & Mixture Controls

Two throttle controls are provided in throttle quadrants, one on each side of the cockpit, giving each pilot engine control. The mixture control is located on the lower section of the main instrument panel just to the right of the starter button. PULL OUT for LEAN, PUSH IN for RICH.

7.4.2 Propeller Pitch Controls

Propeller pitch control levers are provided in the throttle quadrants beside the throttle control levers. Push FORWARD to INCREASE RPM and pull AFT to DECREASE RPM.

7.4.3 Carburettor Heat Control

The carburettor heat control is located on the lower section of the main instrument panel just to the left of the cabin heat control. Pull fully out for HOT, push fully in for COLD.

7.4.4 Engine Starting Controls

T6 — The engine starter is engaged by pressing the starter button situated centrally under the instrument panel. ***Do not engage the starter while the propeller is rotating*** otherwise serious damage may result. The ignition switch is key operated and located on the electrical panel. Each magneto may be individually selected but the engine is normally started on LEFT.

T6/24 — The engine starter is engaged by rotating the combined ignition/starter switch fully clockwise past the spring detent. This position automatically selects LEFT magneto only for the start. ***Do not engage the starter while the propeller is rotating*** otherwise serious damage may result.

7.5 Aircraft Controls

7.5.1 Control Column

The control column is situated forward of the centre armrest and operates the ailerons, flapperons, and the elevator. The elevator is controlled by fore and aft movement of the control column and the ailerons/flapperons by side to side movement.

7.5.2 Rudder Pedals

The rudder pedals are of conventional form and are adjustable over four inches. They are rigidly connected to the rudder, giving positive rudder control. Direct connection from the pedals to the nosewheel provides nosewheel steering.

7.5.3 Trim Controls

T6 — Elevator trim is provided by means of the trim lever protruding from the centre armrest. It operates in the natural sense, ie. its down movement trims the aircraft nose down, up trims nose up. To trim first disengage the lever with the left hand by pulling it to port, operate up or down, until there is no stick force on the control column and re-engage by pushing it home (ie. to starboard).

T6/24 — Elevator trim is provided by an electric actuator controlled by a trim switch mounted on the control column. A trim position indicator is located on the centre lower

panel below the ignition/starter switch.

7.5.4 Flap Controls

T6 — The flap lever is situated on the left hand wall of the cockpit and is accessible to the pilot in the port seat. The flap lever operates in the natural sense and has five locking positions:

- **Zero notch:** Flaps fully down, cruising configuration
- **First notch:** 4° of flap — for reduced speed in the circuit and slow cruising
- **Second notch:** 8° of flap — normally used for take-off
- **Third notch:** 12° of flap — for landing in gusty weather conditions
- **Fourth notch (full):** 15° of flap — for landing, provides steepest rate of descent

T6/24 — The flaps are operated by an electric actuator controlled by two flap selector switches, one on either end of the main instrument panel, providing either pilot with the ability to operate the flaps. A flap position indicator is located on the centre lower panel below the ammeter.

7.5.5 Wheel Brakes

The main wheels are fitted with hydraulic disc type brakes. Even pressure for braking is supplied to both brakes when either hand brake lever is pulled. A parking brake selector of 'lift to park' type is centrally mounted and brakes can be locked ON by first pulling the brake handle and then raising the park selector.

7.6 Flight Instruments

The cockpit is equipped with adequate instruments for purposes of normal flying. The Artificial Horizon and Directional Gyro are operated by suction provided by an engine driven vacuum pump. The Turn and Bank Indicator is electrically powered. The Air Speed Indicator, Vertical Speed Indicator and Altimeter are operated by the pitot static system, comprising a combined pressure and static head on the outboard end under the starboard wing. A magnetic compass is installed centrally on top of the instrument panel.

7.7 General Equipment

7.7.1 Cockpit Canopy

A sliding perspex canopy covers the cockpit. An external handle protruding centrally from the top leading edge of the canopy enables the canopy to be opened by simply turning this through 90° and pulling backwards. The cockpit is entered from each side by climbing on the wing, opening the sliding canopy fully and stepping into the cockpit. The canopy may

be opened in flight up to speeds of 70 KIAS and on the ground during taxying. The canopy should be closed during take-off, climb and landing.

7.7.2 Seats & Harnesses

Non adjustable seats are fitted and equipped with full restraint harnesses.

7.7.3 Maps & Stowage

A stowage space is provided for maps, etc. in the form of flush pockets on the side of the cockpit.

7.7.4 Baggage Compartment

A baggage compartment is situated aft of the seat frame with a structural capacity of 45 kg (100 lbs), and a volumetric capacity of 7.5 ft^3.

7.7.5 Cabin Heating & Ventilation

Cabin heating is provided by warm air from an outlet above the feet at each pilot position. The hot air control is situated centrally under the instrument panel. Pull for cabin heat ON. Demisting of the windscreen is coupled together with cabin heating. To demist the windscreen the cabin heat must be ON. Ventilation is provided through an intake on each side of the cockpit with an individual control at each pilot position.

7.8 Emergency Equipment

7.8.1 Fire Extinguisher

When fitted fire extinguishers are usually mounted on quick release brackets in front of the seats.

7.8.2 First Aid Kits

Carriage of first aid kits is not specifically provided for, however when carried these can be carried in the baggage compartment.

7.8.3 Emergency Exits

When necessary the sliding canopy can be opened rapidly in flight at speeds up to 70 KIAS by unlocking and pulling backwards on the handle in the normal manner.

SECTION 8 — HANDLING, SERVICE & MAINTENANCE

8.1 Introduction

This section contains recommended procedures for proper ground handling and routine care and servicing of your Airtourer aircraft. It also identifies certain inspection and maintenance requirements which fall into the following categories: General & Scheduled Aircraft Maintenance; Alterations or Repairs; and Preventative Maintenance & Minor Field Repairs.

All inspections, maintenance, and repairs must be conducted in accordance with the applicable Civil Aviation Rules in the country of registration of the aircraft. To the extent permitted by the CAR Part 43 Appendix A, preventative maintenance and minor field repairs may be carried out by a suitably licensed pilot.

8.2 Identification Plate

The Serial Number and Registration Marking can be found in the aircraft log books as well as on an identification plate located on the rear face of the pilots seat back.

8.3 Aircraft Documents

The following documents are applicable to the operation and maintenance of this aircraft:

- Pilot's Operating Handbook & Approved Flight Manual
- Airframe Log Book
- Engine Log Book
- Propeller Log Book
- Airtourer Maintenance and Repair Manual
- Airtourer Spare Parts Catalogue
- Lycoming O-320 Engine Operator's Manual

8.4 Aircraft Inspection, Maintenance & Repair

8.4.1 General & Scheduled Aircraft Maintenance

Maintenance shall be carried out in accordance with the Airtourer Maintenance and Repair Manual and other approved maintenance data.

8.4.2 Alterations or Repairs

Alterations and repairs shall be carried out in accordance with the Airtourer Maintenance and Repair Manual. FAA publication AC 43.13-1A&2A or subsequent issues provides additional information relating to alterations and repairs but this information may only be used to the extent authorised by the regulations.

8.4.3 Preventative Maintenance & Minor Field Repairs

Limited maintenance and minor repairs may be carried out by a suitably licensed pilot on an aircraft he owns or operates. The remainder of Section 8 of this manual provides information that will assist the pilot to conduct limited maintenance and minor repairs. It is highly recommended that a copy of the Airtourer Maintenance and Repair Manual is available to the pilot when he performs any maintenance on the aircraft.

8.5 Ground Handling

The aircraft can be easily manoeuvred by hand on the ground by having one man pulling or pushing as required using the tow bar attached to the nose wheel. When required additional people can assist by pushing on the wing (not the control surfaces).

Handgrips, Walkways & Steps — A handgrip is provided on each side of the cockpit just below the sliding canopy. A walkway coated with a black anti-slip material extends over the wings on both sides of the cockpit. It is not permissible to stand or walk on any other part of the wing. ***Do Not Stand on the Flap!***

8.6 Parking

When possible park the aircraft into the anticipated wind and set the parking brakes. Do not set the parking brakes during cold weather when accumulated moisture may freeze the brakes, or when the brakes are very hot. Lock the elevator and aileron controls using the control lock. The rudder may be locked using a suitable locally manufactured external rudder lock between the rudder balance horn and the fin. Install wheel chocks when available. When severe weather conditions or high winds are anticipated, the best precaution is to hangar the aircraft.

8.7 Tying Down

1. Park the aircraft into wind when practical.
2. Lock the ailerons and elevator (and rudder if possible) as described above.
3. Chock the wheels.

4. Attach a strong (recommended at least 400 kg tensile strength) tie down rope to the wing tie-down lugs, and the tail lug at approximately 45° to the ground. Leave some slack to avoid damage due to rope shrinkage that sometimes occurs when a moist rope dries out.

8.8 Jacking

The aircraft may be jacked in order to service the landing gear, change wheels/tyres and to perform other minor service functions:

1. Place the two main jacks under the wing surface aft of the main undercarriage legs. Use suitable jacking adaptors so that the jacks are securely engaged in the jacking point holes.
2. Place a third jack under the jacking point located on the lower firewall just to the right of the nose undercarriage leg.
3. Raise the jacks until the wheels are clear of the ground as required keeping the aircraft relatively level during the jacking process.

Notes:

1. *If only one wheel needs to be raised clear of the ground such as for changing a wheel or tyre, a single jack may be used. Ensure the aircraft is securely chocked to prevent movement when jacked.*
2. *If only one main wheel needs to be raised clear of the ground and a suitable jacking adaptor is available to attach to the main undercarriage leg, a small bottle jack or similar may be used.*

8.9 Levelling

The aircraft is longitudinally level when an accurate level placed along an upper longeron in the cockpit area gives a level indication. The aircraft is laterally level when an accurate level placed across the cockpit between the two upper longerons gives a level indication.

8.10 Fluid Servicing

8.10.1 Fuel System

Filling Fuel Tanks — Observe all the required precautions for handling fuel and filling tanks. Ensure that the aircraft is bonded to EARTH using the earthing point on the nose leg.

Fuel Draining/Sampling — A quick drain fuel drain fitting is provided in the fuel tank sump under the centre fuselage. Fuel should be drained/sampled from this point before the first flight of the day and after each subsequent refuelling.

8.10.2 Engine Lubrication System

Filling the Engine Sump — The engine sump should be filled to the operating level with the lubricating oil specified in para 2.4.3. Open the cowl access door to gain access to the oil dipstick/filler tube.

Draining the Sump — The engine sump is drained by removing the lower engine cowl and removing the sump drain plug. Ensure that the sump drain plug is correctly replaced and locked prior to refilling the engine with oil.

8.11 Wheel Brake Fluid

The brake fluid level may be checked by opening the left hand engine cowl and removing the master cylinder reservoir cap. If necessary it may be topped up with mineral based hydraulic fluid as specified in the Airtourer Maintenance and Repair Manual. *Ensure that no contaminants are allowed to enter the reservoir, particularly when removing or replacing the cap.*

8.12 Undercarriage

The main undercarriage is a simple spring steel leg and apart from normal inspections for security, is maintenance free between scheduled maintenance inspections. The nose undercarriage leg utilises a coil spring as the shock absorbing medium. This spring is damped using a automotive type hydraulic damper. The spring requires no maintenance. The hydraulic damper requires no maintenance in the field.

8.13 Tyres

The tyres should be carefully checked for correct inflation, cuts, abrasions, wear, slippage and other obvious defects and replaced if necessary. Recommended tyre inflation pressures for normal operations:

Wheel	Normal Operations	Maximum (hard surfaces)
Main Wheels	23 psi (159 kPa)	35 psi (241 kPa)
Nose Wheel	18 psi (124 kPa)	25 psi (172 kPa)

8.14 Induction Air Filter

Dust and dirt must be prevented from entering the engine induction system. Dust and dirt ingested into the engine is probably the greatest single cause of premature engine wear. When operating in a very dusty environment the filter may need to be cleaned more regularly than the time periods specified in the Airtourer Maintenance and Repair Manual.

8.15 Battery Service

T6 — The battery is a conventional lead acid aircraft battery. Servicing involves adding distilled water to maintain the electrolyte level, checking and cleaning the terminals, and checking the battery's security. Spilled electrolyte or corrosion may be neutralised and cleaned off using bicarbonate of soda (baking soda) and clean water.

T6/24 — The battery is a nickel cadmium aircraft battery. Maintenance on this battery type must be carried out by suitably trained maintenance personnel only. ***No pilot maintenance is permitted.***

8.16 Flyable Storage

Aircraft in storage for a maximum of 30 days or those which receive only intermittent operational use for the first 50 hours are considered to be in flyable storage status. If the aircraft is not flown the propeller should be rotated by hand through at least six revolutions every seventh day. Stop the propeller 45°–90° from the original position.

> **WARNING:** *For maximum safety, check that both magnetos are OFF, the throttle closed, the mixture control is in the ICO position, and the aircraft is secured before rotating the propeller by hand. Do not stand within the arc of the propeller blades while turning the propeller.*

After 30 days, the aircraft should be flown for at least 30 minutes. If the aircraft is to be out of service for a period longer than 30 days the engine should be inhibited in accordance with the engine manufacturers instructions.

> **NOTE:** *The best method of preventing corrosion of the cylinders and other internal parts of the engine is to fly the aircraft at least once a week for long enough to reach normal operating temperatures. This will vaporise moisture and other by-products of combustion.*

8.17 Cleaning and Similar Care

8.17.1 Windshield & Canopy

The windshield and canopy are made from a plastic perspex material. The following procedure is recommended:

1. Flush with clean water and dislodge excess dirt, mud, etc. using your bare hand so that you can feel any hard to remove lumps and be able to gently remove them.
2. Wash with a mild soap and warm water. Use a soft cloth or sponge. Do not rub excessively.
3. Rinse thoroughly, then dry with a clean moist chamois. Do not rub with a dry cloth as this builds up an electrostatic charge which attracts dust. Oil and grease may be removed by rubbing lightly with a soft cloth moistened with kerosene. *Do not use volatile solvents such as gasoline, alcohol, benzine, carbon tetrachloride, lacquer thinner or most commercial window cleaning sprays, as they will soften and craze the plastic.*
4. After washing, the windshield and canopy should be cleaned using an aircraft windshield cleaner following carefully the manufacturers instructions.

8.17.2 Painted Surfaces

The painted exterior surfaces of the aircraft can be washed using a mild detergent and water. Special aircraft cleaning detergents may be used or alternatively an automotive liquid detergent provided it is non corrosive and contains no abrasive materials. Stubborn oil and grease may be removed using a small amount of solvent such as kerosene.

8.17.3 Propeller Care

Preflight inspection of propeller blades for nicks, and wiping them occasionally with an oily cloth to clean off grass and bug stains will ensure long and trouble free service. Small nicks on the propeller, particularly near the tips and on the rear face near the leading edge and on the leading edge itself, should be dressed out as soon as possible as these nicks produce stress concentrations, and if ignored, may result in cracks and subsequent blade failure. Never use an alkaline cleaner on the blades; remove grease and oil with a suitable solvent such as kerosene.

8.17.4 Engine Compartment

The engine compartment should be kept clean to minimise any danger of fire, and to allow proper inspection of engine components. The engine and engine compartment may be washed down with a suitable solvent, then dried thoroughly.

> **CAUTION:** *Particular care should be given to electrical equipment before cleaning. Solvent should not be allowed to enter magnetos, starter, generator/alternator, and the like. These components should be protected before saturating the engine with solvent. Any oil, fuel, and air openings on the engine and accessories should be covered before*

> *washing the engine with solvent. Caustic cleaning solutions should be used cautiously and should always be properly neutralised after their use.*

SECTION 9 — SUPPLEMENTS

9.1 Introduction

From time to time Flight Manual Supplements may be issued by the aircraft manufacturer. These are usually associated with modifications or the installation of specific equipment. Each supplement contains a brief description, and where applicable, operating limitations, emergency and normal procedures, and the effect on aircraft performance.

Only supplements which are produced by the manufacturer, and listed in para 9.2 below, apply to this individual aircraft. Except for supplements included in the initial issue of the manual all supplements must be individually and formally issued for the aircraft as an amendment to the manual.

> **NOTE:** *When operating in accordance with the provisions of a supplement any limitations and/or procedures contained in that supplement supersede the corresponding limitations and/or procedures contained in the basic Flight Manual.*

9.2 Supplements Applicable to This Aircraft

Reserved.

End of Airtourer T6 — CAA Approved Flight Manual & Pilot's Operating Handbook